NEURALPATH CHEATSHEET

Module 6: Generative AI & Modern LLMs

Learn VAEs, GANs, diffusion & flow matching (DiT), scaling laws, RAG, LoRA/QLoRA fine-tuning, M

Core Formulas, Mental Models & Engineering Rules

- VAE maximizes ELBO = Reconstruction - KL(q(z|x) || p(z)).
- Flow Matching & DiT learn straight velocity fields from noise to data in 4-20 NFE steps.
- RAG: hybrid search (dense + BM25) + reranker + Ragas triad (Faithfulness, Relevance, Recall).
- LoRA/QLoRA: freeze W_0 (4-bit NF4), learn low-rank delta W = B * A.
- Agents & MCP: expose external tools behind scoped, audited JSON-RPC MCP servers.

Estimated study time: 7 hours · Badge: Generative Systems Engineer